A validated enclosure of Robertson's stiff problem on $[0, 4000]$ and the true error of a Rust port of CVODE against it

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Preprint v1, 2026-11-14

doi:10.5281/zenodo.24216837 (not peer reviewed)

Zenodo DOI to be assigned on release (not peer reviewed)

AI-assistance disclosure and audit status. This preprint, the enclosure code, the comparison code and the claim ledger were produced by AI agents (Claude, Anthropic) under the author's direction; the author is responsible for the content. **The result was independently audited once by a different AI agent instance (`data/AUDIT.md`, verdict ACCEPT_WITH_CORRECTIONS), not by a human.** The audit's eleven corrections are applied (Section 8). **No human audit exists.** The preregistration hash is self-attested and was first committed to version control after the production run had started (Section 8).

Abstract

We compute a validated enclosure of Robertson's stiff chemical-kinetics problem ($k = 1/25,\ 10^4,\ 3 \cdot 10^7$, $y(0) = (1, 0, 0)$, exact rational constants) at ten output times from $t = 0.1$ to $t = 4000$, with a hand-written a-posteriori integrator in 160-bit Arb ball arithmetic (Sage 9.5). **The enclosure is rigorous under stated assumptions** (Arb is correct; the log-norm lemma, proved here in prose, is not formalised): the required relative width of 10^{-6} is met everywhere and the reported widths are far smaller (at most $6 \cdot 10^{-10}$, in y_2 at $t = 4000$). **It was independently re-derived by the auditor's different method (Lohner interval Taylor in MPFI) at $t = 1, 10, 40$ only**; for $40 < t \leq 4000$ it rests on one implementation plus a code audit. Against the enclosure we measure the true global error of the `cvode` crate of rusty-SUNDIALS (a Rust port, *not* C CVODE, which is not installed on the host) for rtol $= 10^{-4}, \ldots, 10^{-10}$. For y_2 the apparent absence of any tolerance violation is only because atol dominates the tolerance: measured against rtol $|y_2|$ the error is up to 169× larger. We also note that a "provable violation" compares a global error with a local-error tolerance. No claim is made about general stiff problems, $t \geq 10^4$, or C CVODE. Producer and one independent audit are AI agents; no human audit; no Lean.

Markers. [L]: a cited source, taken as stated. [C]: executed code with an argument written down but not machine-checked (no Lean here). [X]: a recorded absence, an assumption, or a list of what was not done, supporting nothing by itself. There is no Tier A claim in this paper; we never write "formally verified". Identifiers `PH2-·-nnnn` refer to `ledger/ledger.json`.

1 Problem and conventions

Robertson's problem is

$$\begin{aligned} \dot{y}_1 &= -k_1 y_1 + k_2 y_2 y_3, \\ \dot{y}_2 &= k_1 y_1 - k_2 y_2 y_3 - k_3 y_2^2, \\ \dot{y}_3 &= k_3 y_2^2, \qquad y(0) = (1, 0, 0), \end{aligned} \tag{1}$$

with $k_1 = 1/25$, $k_2 = 10^4$, $k_3 = 3 \cdot 10^7$, all exact rationals. Since $\dot{y}_1 + \dot{y}_2 + \dot{y}_3 = 0$ identically, $y_1 = 1 - y_2 - y_3$ holds exactly and we integrate the planar system for $(a, b) = (y_2, y_3)$,

$$\dot{a} = k_1(1 - a - b) - k_2 ab - k_3 a^2, \qquad \dot{b} = k_3 a^2, \qquad (a, b)(0) = (0, 0), \tag{2}$$

a polynomial field f of degree two. The reduction is an equivalence, not an assumption. The Jacobian $J(a, b) = \begin{pmatrix} -k_1 - k_2 b - 2k_3 a & -k_1 - k_2 a \\ 2k_3 a & 0 \end{pmatrix}$ is *affine* in (a, b). An *enclosure at time* t is a box $I_1 \times I_2 \times I_3$ of ball intervals containing the unique solution $y(t)$. The output times are $t \in \{0.1, 0.4, 1, 4, 10, 40, 100, 400, 1000, 4000\}$ (the preregistered list also had 10^4 and $4 \cdot 10^4$, which were not reached). The times 0.1 and 0.4 are the *doubles* handed to the Rust solver, converted to exact rationals, not 1/10 and 2/5. The goal set in advance was a relative width $\leq 10^{-6}$ in every component at every output time, up to $T = 40000$.

2 The method, as the exact theorem the code relies on

The integrator is a validated one-step method in the family of a-posteriori (logarithmic-norm) enclosures; no novelty is claimed for the lemma. We state precisely what the code uses.

Setting of one step. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be a polynomial vector field of degree ≤ 2. On a step $[0, h]$ let z be the true solution with $z(0) \in \mathbb{R}^2$, and let q be a *polynomial* (the reference; degree $p - 1$ with dyadic coefficients, $p = 24$ in the production run) with $q(0) = m$. Let P be a constant invertible 2×2 matrix and $\|v\|_P := \|P^{-1} v\|_2$. Assume $\|z(0) - m\|_P \leq \varepsilon_0$. Define the defect $D(t) := f(q(t)) - q'(t)$, which is a polynomial of degree $\leq 2p - 2$ in t with coefficients D_k, and

$$\bar{d} := \sum_{k=0}^{2p-2} \|P^{-1} D_k\|_2 \, h^k \quad \geq \quad \sup_{t \in [0,h]} \|P^{-1} D(t)\|_2$$

(triangle inequality and $t^k \leq h^k$; no Horner evaluation on intervals, no a-priori tube needed). Let $\rho > 0$ and let $U \subset \mathbb{R}^2$ be a box such that for every $t \in [0, h]$ and every w with $\|w\|_2 \leq \rho$, the whole segment $[q(t),\, q(t) + Pw]$ lies in U. Let

$$\nu \;\geq\; \sup_{y \in U} \; \lambda_{\max}\Big(\tfrac{1}{2} \big(P^{-1} J(y) P + (P^{-1} J(y) P)^{\top} \big) \Big).$$

Theorem 1. *If* $(\varepsilon_0 + \bar{d}\, h)\, e^{\max(\nu, 0) h} < \rho$, *then* z *exists on* $[0, h]$ *and for all* $t \in [0, h]$

$$\|z(t) - q(t)\|_P \;\leq\; \varepsilon_0 e^{\nu t} + \bar{d} \int_0^t e^{\nu s}\, ds \;<\; \rho.$$

Proof. Put $e = z - q$ and $w = P^{-1} e$, so $w' = P^{-1}(f(z) - f(q)) + P^{-1} D$ as long as z exists. *The key identity:* because f is quadratic, J is affine, hence

$$f(z) - f(q) = \int_0^1 J(q + s\, e)\, e \, ds = J\big(q + \tfrac{1}{2} e\big)\, e \quad \text{exactly,}$$

with no remainder term (the integrand is affine in s and the midpoint rule is exact for affine functions). Thus $w' = \big(P^{-1} J(q + e/2) P\big) w + P^{-1} D$. While $\|w\|_2 \leq \rho$, the points $q(t) + se$, $s \in [0, 1]$, lie in U, so $\langle w, P^{-1} J(q + e/2) P w \rangle \leq \nu \|w\|^2$ and the upper Dini derivative satisfies $D^+ \|w\| \leq \nu \|w\| + \bar{d}$. The comparison principle gives $\|w(t)\| \leq g(t) := \varepsilon_0 e^{\nu t} + \bar{d} \int_0^t e^{\nu s} ds$ for as long as the tube condition holds. Moreover $g(t) \leq (\varepsilon_0 + \bar{d} t) e^{\max(\nu, 0) t} < \rho$ on $[0, h]$. Suppose z left the closed tube $\{\|w\| \leq \rho\}$ or ceased

to exist; let $t^* \in (0, h]$ be the first time with $\|w(t^*)\| = \rho$ (the solution is continuous and cannot escape without touching the boundary). On $[0, t^*]$ the tube condition holds, so $\rho = \|w(t^*)\| \le g(t^*) < \rho$, a contradiction. Hence z stays in a compact set, extends to all of $[0, h]$, and the bound holds. □

Remark 1 (Why this is valid for the nonlinear system). The one-sided Lipschitz constant ν is taken over the *whole* box U containing the tube and every segment between z and q, not at q alone, and the identity $f(z) - f(q) = J(q + e/2)e$ places the relevant Jacobian inside that box. Nothing in the argument is linear. The defect is an *exact* polynomial: the reference q is a polynomial, $f(q)$ is a polynomial, so D has no truncation remainder; its coefficients come from exact ball Cauchy products of the dyadic coefficients of q.

Chaining steps. After a step to the end time, put $m' := \mathrm{mid}(q(h))$ (a dyadic rational), and let $\mathrm{jump} := q(h) - m'$ (a ball). With the next step's matrix P' and $\varepsilon_{\mathrm{end}}$ the bound of Theorem 1 at $t = h$,

$$\|z(h) - m'\|_{P'} \;\le\; \sigma_{\max}\big(P'^{-1}P\big)\,\varepsilon_{\mathrm{end}} + \|P'^{-1}\,\mathrm{jump}\|_2 =: \varepsilon_0'.$$

The code uses the exact spectral norm of a 2×2 matrix, $\sigma_{\max}^2 = \big(F^2 + \sqrt{F^4 - 4\det^2}\big)/2$ with F the Frobenius norm, rather than the preregistered Frobenius norm itself (deviation D5: valid, tighter). The initial value is exact, $\varepsilon_0 = 0$. At an output time the components obey $|z_i - q_i| \le \|\mathrm{row}_i(P)\|_2\,\varepsilon_{\mathrm{end}}$ by Cauchy–Schwarz; the output times are hit exactly by truncating the last step, and y_1 is the ball $1 - I_2 - I_3$.

What is *not* used. Positivity and invariance of the simplex are not used (the region is all of $\mathbb{R}^2$; the preregistration listed an invariance-based restriction, deviation D4, which would only have shrunk the region and is not needed).

Step control. A step of length h is accepted only if $\bar{d}\,h \le \tau = 10^{-22}$ and the bootstrap condition holds; otherwise h is halved. The basis P is chosen per step from floating-point eigenvectors of J at $q(h/2)$ (real, or real and imaginary parts of a complex eigenvector, normalised columns), with the identity as fallback. These choices affect only efficiency, never validity.

3 Implementation and ball arithmetic

The single file `code_producer/ph2_enclose.py` (about 370 lines) uses Sage 9.5 `RealBallField(160)` (Arb). Every inequality that enters the theorem is evaluated in balls and checked with `.upper()` / `.lower()`: $\bar{d}$ by ball Cauchy products and `up()` (round upward); ν by the closed form $\lambda_{\max} = \frac{a+c}{2} + \sqrt{(\frac{a-c}{2})^2 + b^2}$ of the symmetric part of $P^{-1}JP$, evaluated in balls with J the interval Jacobian over U (exact enclosure, as J is affine); the bootstrap test `emax.upper() < rho.lower()`; and $\varepsilon_{\mathrm{end}}$ rounded up.

Where floating point occurs, and why it is safe. (i) The choice of h, converted to an exact rational. (ii) The eigenvector basis P: doubles are converted *exactly* to balls, P^{-1} is the ball adjugate formula, and a determinant ball containing 0 is rejected. (iii) The comparison with τ for step control. (iv) Display. The reference-polynomial coefficients are exact dyadic rationals taken from midpoints of a ball recursion; "any source is allowed, only the checks matter". The helpers `sq` and `rsqrt` clip lower ends at 0, valid because the arguments are mathematically nonnegative. Doubles enter balls through `as_integer_ratio`, which is exact.

Resumability. The production run was interrupted by session ends and reboots and resumed from checkpoints that store t, h, m, P as exact dyadic rationals, ε_{end} as its exact upper bound, and the jump as exact midpoint plus a radius added outward; any resumed state is therefore a valid enclosure state. The auditor's resume test (a run to $t = 0.4$ interrupted twice) reproduced an uninterrupted run bitwise; this covers $t \le 0.4$ only, and for the resumes after $t \approx 1000$ the structural argument applies. A crash-only edge case (a kill between the last step onto an output time and the output save) makes the next resume raise a `NameError`: visible, not silent.

4 Controls

Four controls were preregistered (`code_producer/controls.py`); results in `logs/controls.json`. *Positive.* C+1, the stiff linear system $y' = Ay$, $A = \left(\begin{smallmatrix} -1 & 1 \\ 0 & -10^4 \end{smallmatrix}\right)$, $y(0) = (1, 1)$ with the exact solution known in balls: contained at $t = 0.1, 1, 10$ at order $p = 24$ and $p = 6$. C+2, the logistic equation $y' = y(1 - y)$ with companion $w' = -2w$ (exact solution $1/(1 + 9e^{-t})$): contained at $t = 1, 5, 10$. *Negative.* C-2, the same low order $p = 6$ with the defect term dropped (deliberately unsound): containment is lost at every time for both systems, so the controls have power against a missing defect. C-1, the exact value shifted by the factor $1 + 10^{-6}$, must be rejected.

Adjudication. C-1 is rejected for the linear y_1 and for both logistic components at $p = 24$, but is *never* rejected for the linear $y_2 = e^{-10^4 t}$ (at $t = 0.1$ this is e^{-1000}; the enclosure's relative width is of order $3 \cdot 10^7$, so a 10^{-6} relative shift is far below any representable absolute width). The literal preregistered kill rule would trigger ("C-1 not rejected ⇒ BLOCKED"); the auditor adjudicated that the test has *no power* for that component, which is not a soundness failure. In the Robertson run itself y_2 has relative width at most $6 \cdot 10^{-10}$ at $y_2 \sim 10^{-6}$ to 10^{-5}, so a 10^{-6} relative error would be rejected. At $p = 6$ most shifts are not rejected because the low-order boxes are wider than 10^{-6} (only the logistic y at $t = 10$ is).

Code version. `logs/controls.json` was produced by an earlier version of `ph2_enclose.py` (controls at 12:52; the final file is dated 13:34 on 2026-11-11). The auditor reran the controls on the final code (`code_auditor/ctrl_rerun.log`) with identical containment and broken-control verdicts; relative widths differ only around the 8th digit, for the linear y_2 and the linear y_1 at $t = 10$. The auditor's rerun is the control evidence for the production code. [C]

5 Results: the enclosure

Table 1 gives the midpoints of the ten boxes, and Table 2 their relative widths. All boxes are exact rationals in `enclosures.json`; the display shows rounded midpoints. The run stopped on its CPU budget (cumulative 29 900 s = 8.31 core-hours, 2 850 083 accepted steps, 4 389 852 attempts; last log line `stopped: cpu_cap` at $t \approx 4007.8$), not through a failure of the method. The preregistered outcome rule therefore gives **PARTIAL**: relative width $\le 10^{-6}$ holds up to $T = 4000$; $T = 40000$ was not reached. The largest relative width is $5.99 \cdot 10^{-10}$ (y_2, $t = 4000$), at least 1600× below the requirement. [C]

Step history (a subsample). The checkpoint stores $(t, h, \nu, \varepsilon_{\text{end}})$ every 20th step, and RESULT.md lists every 20th of those; we show four rows of that subsample: $t = 268$: $h = 1.9 \cdot 10^{-3}$, $\nu = 1.2 \cdot 10^{-3}$, $\varepsilon_{\text{end}} = 7.0 \cdot 10^{-18}$; $t = 1014$: $1.8 \cdot 10^{-3}$, $9.3 \cdot 10^{-5}$, $8.0 \cdot 10^{-17}$; $t = 2127$: $1.4 \cdot 10^{-3}$, $-1.8 \cdot 10^{-4}$, $2.8 \cdot 10^{-16}$; $t = 4008$: $9.7 \cdot 10^{-4}$, $-2.0 \cdot 10^{-4}$, $2.7 \cdot 10^{-16}$. The step size stays near 10^{-3} throughout, set

Table 1: Midpoints of the enclosure boxes ($y_1 = 1 - y_2 - y_3$ exactly). The last column marks the output times at which the auditor's independent integrator (Section 6) also produced a box; "–" means no independent box exists.

t	y_1	y_2	y_3	independent box
0.1	0.996077747442	3.580437235e-05	0.003886448185	–
0.4	0.985172113861	3.386395379e-05	0.014794022185	–
1	0.966459737333	3.074626579e-05	0.033509516401	yes
4	0.905518678584	2.240475688e-05	0.094458916659	–
10	0.841369923841	1.623390938e-05	0.158613842249	yes
40	0.715827068719	9.185534765e-06	0.284163745746	yes
100	0.617234882396	6.153591275e-06	0.382758964013	–
400	0.450518668471	3.222901442e-06	0.549478108627	–
1000	0.336874530661	2.013702318e-06	0.663123455637	–
4000	0.183202257777	8.942371253e-07	0.816796847986	–

Table 2: Relative widths of the boxes and the number of accepted steps up to each output time.

t	rel. width y_1	rel. width y_2	rel. width y_3	steps
0.1	1.28×10^{-21}	1.31×10^{-17}	2.06×10^{-19}	71
0.4	2.41×10^{-21}	2.64×10^{-17}	1.00×10^{-19}	136
1	9.56×10^{-21}	1.17×10^{-16}	1.68×10^{-19}	265
4	1.31×10^{-19}	2.30×10^{-15}	7.10×10^{-19}	928
10	8.19×10^{-19}	1.99×10^{-14}	2.31×10^{-18}	2 372
40	7.84×10^{-18}	3.02×10^{-13}	1.00×10^{-17}	11 382
100	2.31×10^{-17}	1.16×10^{-12}	1.87×10^{-17}	34 219
400	7.52×10^{-17}	5.25×10^{-12}	3.08×10^{-17}	185 578
1000	9.41×10^{-16}	7.87×10^{-11}	2.39×10^{-16}	560 307
4000	5.84×10^{-15}	5.99×10^{-10}	6.55×10^{-16}	2 843 424

by the defect tolerance τ rather than by stability; the error bound ε_{end} grows slowly to about $3 \cdot 10^{-16}$ (around $t \approx 2400$) and then decreases slightly ($\nu < 0$ from $t \approx 1345$ on).

Assumptions behind the enclosure. (1) Arb/Sage 9.5 ball arithmetic is correct. (2) The step theorem (Section 2) is argued in prose, not formalised. (3) The code implements the theorem; this was checked by the auditor's line-by-line read (obligations defect, one-sided Lipschitz, bootstrap, jump and basis change, output, rounding, resume: all discharged, the rounding item with the caveat that rigor rests on Arb).

6 Independent check

The auditor wrote a different integrator without reading the producer's step logic: a Lohner interval Taylor method of order 18 with an a-priori enclosure by Moore's lemma, a Lagrange remainder, the mean-value form with forward-mode variational coefficients and a QR-reorthogonalised parallelepiped, in MPFI intervals (128 bits), a different library from Arb and without any logarithmic norm. It reached $t = 40$ (263 061 steps, 1 360 s). At $t = 1, 10, 40$ **its boxes lie inside the producer's on all three components**, the midpoints agree to $1 \cdot 10^{-26}$ (at $t = 1$, y_1, y_3), $4 \cdot 10^{-27}$ (at $t = 10$) and $1.6 \cdot 10^{-27}$ (at $t = 40$, y_1, y_3; $9 \cdot 10^{-30}$ for y_2), and the auditor's radii are about 10^3 to 10^8 times narrower (at $t = 40$: $2.4 \cdot 10^{-25}$ against $2.8 \cdot 10^{-18}$). At $t = 10$ for y_2 the producer's midpoint lies

outside the auditor's much narrower box; this is consistent, since the auditor's box is inside the producer's. The boxes at $t = 0.1, 0.4, 4$ were *not* compared. [C]

Non-rigorous cross-check at $t = 4000$. scipy Radau in double precision (rtol/atol $10^{-12}/10^{-20}$ and $10^{-13}/10^{-22}$) gives $y(4000) = (0.18320225777671084,\ 8.942371252775939 \cdot 10^{-7},\ 0.8167968479861689)$ at the tighter setting, against the producer's midpoints $(0.1832022577767093,\ 8.942371252775910 \cdot 10^{-7},\ 0.8167968479861654)$. The two agree to about $8 \cdot 10^{-15}$ relative in y_1 and $4 \cdot 10^{-15}$ in y_3. **This is outside the producer's y_1 and y_3 boxes** (radius about $5 \cdot 10^{-16}$); the discrepancy is explained by double-precision roundoff, since the two Radau runs themselves differ by $7 \cdot 10^{-16}$ to $3 \cdot 10^{-15}$ at their own tolerances, and the Radau value is not a validated quantity. Only y_2 falls inside the producer's box (relative width $6 \cdot 10^{-10}$). The cross-check is consistent to about 10^{-14} relative and *cannot discriminate at the box's resolution*; it tests nothing about the validity of the box. [X] **For $40 < t \leq 4000$ no independent rigorous check exists.**

7 The Rust port of CVODE against the enclosure

Solver and settings. The solver is the `cvode` crate of rusty-SUNDIALS (a Rust port of CVODE; BDF, Newton iteration, dense LU, analytic Jacobian, scalar absolute tolerance, `Task::Normal`, one solve through the output times in order), repository branch `merge-mcp-temp`, HEAD `af4886f2`, linked through a Cargo *path* dependency (so the binary's source revision is recorded but not enforced). **There is no C CVODE on this host; nothing here is a statement about the LLNL C library.** The right-hand side uses $\mathrm{fl}(0.04) \neq 1/25$ (relative difference $\sim 10^{-17}$) and double-precision products; we count this as the solver's own arithmetic, i.e. as solver error. The doubles returned by the solver are converted exactly to balls, and the error interval is the ball difference $E_i = |y_{\mathrm{solver,i}} - Y_i|$ with Y_i the enclosure, so the enclosure's width is part of the interval (it is negligible here).

Tolerances. rtol $= 10^{-4}, 10^{-6}, 10^{-8}$ with atol $= 10^{-3}$rtol; all returned output at all twelve requested times. For rtol $= 10^{-10}$ the preregistered atol $= 10^{-13}$ run failed with `ErrTestFailure` at the first output time and produced no output; **the 10^{-10} rows come from an extra, non-preregistered run with** atol $= 10^{-10}$ (deviation D1). Only ten of the twelve times have an enclosure, so ten are compared. The ratio is $E_i/(\mathrm{rtol}|y_i| + \mathrm{atol})$; the flag "provably exceeds" means $\min E_i > \mathrm{rtol} \max |y_i| + \mathrm{atol}$ and "provably within" means $\max E_i \leq \mathrm{rtol} \min |y_i| + \mathrm{atol}$.

Global versus local. The flag "provably > tol" compares the *global* error with a per-component tolerance scale. CVODE controls a *local* weighted-RMS error estimate, so these rows document the size of the global error, not a breach of the solver's contract; that global error can exceed the tolerance and grow with the integration length is textbook for BDF codes [6, 7].

Table 3: Largest global error over the ten output times, in units of rtol $|y_i|$ (upper end of the error interval), per component; last column: largest E_2/atol for y_2. The 10^{-10} row is the non-preregistered atol $= 10^{-10}$ run. Computed by `code_writer/analyse.py` from `comparison.json`.

rtol	atol	y_1	y_2	y_3	y_2: E_2/atol
10^{-4}	10^{-7}	2.71	4.14	2.17	0.019
10^{-6}	10^{-9}	8.1	10.9	8.15	0.091
10^{-8}	10^{-11}	16.6	20.3	13.8	0.16
10^{-10}	10^{-10}	136	169	486	0.0025

Results. Table 3 summarises all 120 error intervals. For y_1 and y_3 the global error exceeds rtol $|y|$ by factors of 2 to 3 (10^{-4}) up to 136–486 (10^{-10}, where the early $y_3 \approx 4 \cdot 10^{-3}$ at $t = 0.1$ is small), and the provable-exceedance count grows as the tolerance tightens (e.g. y_3 at rtol $= 10^{-8}$: 10 of 10 output times). The errors of y_1 and y_3 are equal to three digits at most times (e.g. $2.24 \cdot 10^{-7}$ each at $t = 0.1$, rtol $= 10^{-4}$), as expected when a linear invariant is preserved and y_2 is tiny. Table 4 lists relative errors $E_i/|y_i|$ at four times. **The finding for y_2.** The preregistered flag reports "no provable violation" for y_2 at every tolerance and time, and the first-exceedance table of RESULT.md reads "none" for y_2. That is *only* because the tolerance for $y_2 \leq 3.6 \cdot 10^{-5}$ is atol-dominated ($E_2/\text{atol} \leq 0.16$ throughout). Measured against rtol $|y_2|$ alone, the error of the port exceeds the relative tolerance by up to 4.1× (10^{-4}, $t = 1000$), 11× (10^{-6}, $t = 400$), 20× (10^{-8}, $t = 4000$) and 169× (10^{-10}, $t = 4000$). So an atol-dominated "within tolerance" certificate hides a relative error far above rtol in the small component, which is the one that drives the chemistry. [C] (for this port, these settings, this problem)

Table 4: Relative global error $E_i/|y_i|$ of the Rust port at four output times (upper end of the interval). rtol 10^{-10} uses atol $= 10^{-10}$ (non-preregistered).

rtol	t	y_1	y_2	y_3
10^{-4}	1	3.43e-06	1.76e-05	9.90e-05
10^{-4}	40	4.83e-05	1.58e-04	1.22e-04
10^{-4}	1000	2.71e-04	4.14e-04	1.37e-04
10^{-4}	4000	1.48e-04	6.28e-05	3.32e-05
10^{-6}	1	1.84e-07	8.41e-07	5.32e-06
10^{-6}	40	3.24e-06	9.91e-06	8.15e-06
10^{-6}	1000	4.74e-06	7.07e-06	2.41e-06
10^{-6}	4000	8.10e-06	9.88e-06	1.82e-06
10^{-8}	1	2.20e-09	1.09e-08	6.35e-08
10^{-8}	40	5.47e-08	1.70e-07	1.38e-07
10^{-8}	1000	1.25e-07	1.87e-07	6.36e-08
10^{-8}	4000	1.66e-07	2.03e-07	3.72e-08
10^{-10}	1	1.28e-09	6.69e-09	3.69e-08
10^{-10}	40	4.08e-09	1.26e-08	1.03e-08
10^{-10}	1000	5.91e-09	8.85e-09	3.00e-09
10^{-10}	4000	1.36e-08	1.69e-08	3.05e-09

Recalled reference values. The first-exceedance times by component are in `RESULT.md`. The SUNDIALS example `cvRoberts_dns` prints y at $t = 0.4, 4, 40$ to five digits; we recalled these numbers (they were *not* fetched). That example runs at rtol $= 10^{-4}$ with a vector atol, so the printed values are *solver output*, not the true solution. Their differences from the enclosure midpoints (relative 10^{-6} to $5.5 \cdot 10^{-5}$) are consistent with solver error at rtol $= 10^{-4}$; we draw no conclusion from them. [X]

8 Deviations, preregistration anchoring and audit status

Deviations from `PREREG_ITEM.md` (all disclosed, none blocks): **D1** the non-preregistered rtol $= 10^{-10}$, atol $= 10^{-10}$ run replaces the failed preregistered one (Section 7); **D2** the driver's CPU cap was raised by the main session at several relaunches (cumulative caps recorded as 21600, 11000, 27700 and 29900 s); the total, 8.31 core-hours, is within the preregistered 10; CPU between the last checkpoint and each kill is not in the ledger, which therefore slightly undercounts; **D3** three relaunches followed session ends or reboots (resumption cannot change results, Section 3); **D4** the invariance-based region restriction (O1) was not used, which is the safe direction; **D5** spectral instead of Frobenius

norm in the basis change; **D6** the controls ran on an earlier version of the code (rerun by the auditor with identical verdicts); **D7** the "second implementation or second precision" mentioned in the preregistration draft was not done by the producer; the auditor supplies it up to $t = 40$ only. Further bookkeeping defect: `RESULT.md` originally said "stopped: None" because `run()` sets `res["stopped"]` after `save()` (`ph2_enclose.py`, lines 342–343), so the checkpoint never records it; the true state is `stopped: cpu_cap`. The audit also asked for this code line to be moved; we *deliberately left the code unchanged* so that it is byte-identical to the code that produced the run (its hash is recorded in `result.json`) and corrected only the text.

Anchoring caveat. `PREREG_ITEM.md` was first committed to git two days after the production launch (2026-11-13, commit `00328e1`; launch 2026-11-11 13:34). Its freeze rests on file modification times (11:11) and a self-hash (`PREREG_ITEM.md.sha256`, which matches). `logs/cvode_raw.jsonl` has mtime 13:36, after the freeze; that fits the appended non-preregistered run, but the preregistration's claim that the solver outputs predate the freeze cannot be verified from version control.

Audit status. One independent AI audit (Opus agent, producer $\neq$ verifier; `data/AUDIT.md`), verdict ACCEPT_WITH_CORRECTIONS; eleven corrections applied to `RESULT.md` (correction 8 as in the addendum: "independent check exists for $t \leq 40$"). `result.json` records the hash of `RESULT.md` *before* the corrections; this is the only recorded hash that no longer matches (`data/logs/result_json_hash_check.txt`).

9 Limitations

- **Not C CVODE.** The compared solver is the Rust port; no statement is made about LLNL CVODE.
- **Provenance of the Rust binary.** Cargo uses a path dependency; the source revision (HEAD `af4886f2`) is recorded, not pinned, so that the binary was built from it is unverified. [X]
- **Arb is assumed correct.** The only second implementation (MPFI) covers $t \leq 40$. [X]
- **The lemma is proved in prose, not formalised**; there is no Lean in this work and we do not claim "formally verified". [X]
- $40 < t \leq 4000$ has no independent rigorous check; the Radau comparison cannot test the box. [X]
- **Scope.** One problem (Robertson, these constants, this initial value). No claim for general stiff problems, for $t \geq 10^4$ or $t = 10^{11}$, or for other solver settings (vector atol, other Jacobians).
- **Recalled numbers.** The `cvRoberts_dns` values were recalled, not fetched, and are solver output.
- **Cost.** 8.3 core-hours for $T = 4000$; the method does not scale to the long times (up to $4 \cdot 10^{10}$) at which stiff solvers are routinely run on this problem.
- **AI-produced.** Producer and the one independent audit are AI agents; no human audit.

10 Related work

All statements are relative to a limited search (two web and two alphaXiv queries at scouting, repeated at audit; 2026-11-11 and 2026-11-14), and are "to our knowledge" only. **Validated stiff integration.**

The Nedialkov–Jackson overview of validated interval-Taylor and Hermite–Obreschkoff methods treats the wrapping effect as a stability question and notes that truncation-error formulae make effective validated methods for stiff problems hard [1] (read at snippet/abstract level only). Neumaier's logarithmic-norm enclosures [2] aim at removing the step restriction of Lohner-type methods on stiff systems (secondary: not read in full by us); our theorem is of that type. Sandretto and Chapoutot's DynIBEX provides validated Runge–Kutta methods including implicit Radau and singly-implicit schemes, with a stiff test of oil-reservoir type [3] (abstract read). Generic Taylor-tube validated integration is the subject of arXiv:2604.19068 [4] (listed by the search, not read). VNODE-LP, CAPD, COSY and Ariadne papers, and Flow* and VSPODE-type evaluations on kinetics models mentioned by the auditor's search, were *not individually checked*. **Reference solutions.** The Bari stiff test set [5] gives reference solutions for Robertson computed with the RADAU code in double precision at tight tolerance (secondary reports: not validated). **CVODE-type error.** That BDF/CVODE global error is not proportional to the tolerance and that error control is local is textbook [6, 7]. Our earlier study of Yukawa crystals used CVODE BDF as a relaxer, not compared with an enclosure [8]. The repository's own rusty-SUNDIALS manuscript reports agreement with the C reference on benchmark problems, which is solver-against-solver agreement and not a comparison with the truth (internal; not cited as a source). **Novelty.** To our knowledge no validated enclosure of Robertson's problem to $t \geq 4000$ and no study of CVODE or a port of it against such an enclosure was found. A search is not proof of absence, we make no priority claim, and the authors of the cited works were not contacted. [X]

11 Reproducibility and ledger

Everything is in the bundle: `data/` (lane records with the corrected `RESULT.md`, `AUDIT.md`, preregistration, prior-art screen, `enclosures.json`, `comparison.json`, `cvode_runs.json`, producer code, the Rust runner source, logs, the auditor's scripts and outputs, the writer's scripts) and `ledger/` (claims, evidence blobs, the verbatim Elenchus gate output). The 13 MB checkpoint `logs/enclosure_ckpt.json` is not shipped: only its sha-256 (`data/logs/enclosure_ckpt.json.SHA256.txt`) is stored. The ledger files the enclosure at $t = 0.1, \ldots, 4000$ (independently confirmed at $t = 1, 10, 40$ only) and the true-error tables of the Rust port at Tier C; the independent confirmation for $40 < t \leq 4000$, the provenance of the Rust binary, the correctness of Arb, and "formally verified" (which is *not* claimed) at Tier X; and the cited literature at Tier L. The gate is a bookkeeping check; it licenses nothing.

References

[1] N. S. Nedialkov and K. R. Jackson, overview of validated methods for initial value problems for ODEs (read at snippet/abstract level only; full bibliographic data not recorded).

[2] A. Neumaier, notes on enclosures for initial value problems and stiff systems, `arnold-neumaier.at/ms/ode.pdf` and `ivpBounds.pdf` (secondary; not read in full).

[3] J. Alexandre dit Sandretto and A. Chapoutot, validated Runge–Kutta methods in DynIBEX, arXiv:1804.04847, and the related paper in *Reliable Computing* 22 (abstracts read; entry written from the search record, details not verified).

[4] The Taylor tube method for validated initial value problems, arXiv:2604.19068 (2026; listed by an alphaXiv search, not read).

[5] F. Mazzia and C. Magherini, Test Set for IVP Solvers, `archimede.uniba.it/~testset` (secondary reports only).

[6] E. Hairer and G. Wanner, *Solving Ordinary Differential Equations II*, Section IV.8 (textbook; cited from the prior-art screen, not re-read here; entry from memory).

[7] SUNDIALS/CVODE documentation, on local error control without a global guarantee (cited from the prior-art screen; not re-read here).

[8] X. Callens, Yukawa-crystal preprint on an exact-potential CVODE consistency study, doi:10.5281/zenodo.23293429.